A Rigorous $\mathcal{O}(N^2)$ Quadrature Methodology for Sub-grid Cloud Condensate Integration

Methodology Documentation

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Abstract

Numerical integration of sub-grid scale (SGS) cloud condensate requires resolving a 3D probability space: a 1D vertical layer depth representing the grid-cell geometry, and 2D local turbulence in thermodynamics. Standard heuristic methods, such as total variance projection, fail catastrophically due to the exponential, discontinuous nature of the Clausius-Clapeyron saturation threshold, which causes non-physical hallucinated tails to heavily bias the integral. Conversely, full 3D tensor-product quadrature scales as $\mathcal{O}(N^3)$, proving computationally prohibitive for high-resolution atmospheric models.

This document presents a strictly derived $\mathcal{O}(N^2)$ 2D quadrature method that perfectly preserves the compact support of the vertical path and the cross-covariance of the turbulence. By employing a conditional Rosenblatt transformation, we reduce the problem to an analytical 1D convolution. To achieve $\mathcal{O}(1)$ runtime evaluation, we derive a suite of solvers, including a Logistic-Warped Householder (4th-order) root-finder and Spectral Barycentric rational expansions, providing machine-precision integration suitable for GPU and high-performance computing architectures.

1 Physical Motivation and Problem Formulation

The expected value of cloud condensate q_c within a grid cell of height H is defined by the integral of the saturation excess over the joint probability density function (PDF) of thermodynamic variables, typically liquid water potential temperature T and total water specific humidity q.

Let the mean state at the grid-cell center be $\boldsymbol{\mu} = (\mu_T, \mu_q)^T$, with constant vertical gradients $\nabla\phi = (dT/dz, dq/dz)^T$. The SGS turbulence is parameterized as a bivariate Gaussian with covariance Σ_{turb}. The grid-cell integration requires evaluating:

$$\overline{q_c} = \frac{1}{H}\int_{-H/2}^{H/2}\iint \max(0, q - q_{sat}(T))\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu} + z\nabla\phi, \Sigma_{turb})\, dT\, dq\, dz \tag{1}$$

This represents the integral over a 3D "Pill" distribution: a 2D Gaussian convolved along a 1D rigid line segment of length $L = H\|\nabla\phi\|$.

2 The Failure of Naive Dimensional Reduction

To avoid the prohibitive $\mathcal{O}(N^3)$ computational cost of a full 3D Tensor-Z quadrature, standard models attempt to collapse the 3D integral into a 2D space. These historical approaches introduce severe physical errors.

2.1 Zero-Gradient Baseline (Turbulence Only)

The simplest approach ignores the vertical gradient entirely ($L = 0$), evaluating only Σ_{turb} at the cell center. This severely underestimates condensate by failing to capture saturation triggered at the cooler vertical boundaries of the grid cell.

2.2 Variance Moment-Matching (Total Variance Projection)

A common heuristic attempts to moment-match the total variance by adding the variance of the uniform vertical path to the turbulence:

$$\Sigma_{tot} = \Sigma_{turb} + \frac{H^2}{12}(\nabla\phi)(\nabla\phi)^T \tag{2}$$

The Topological Catastrophe: A uniform path has rigid boundaries (compact support), whereas a Gaussian has infinite soft tails. Utilizing mapped Gauss-Legendre quadrature on Σ_{tot} places integration nodes deep into hallucinated tails. Because $q_{sat}(T)$ decays exponentially with temperature, sampling these non-physical cold tails leads to massive, systemic condensate overestimation.

3 Exact Dimensional Reduction via Conditional Shear

To achieve $\mathcal{O}(N^2)$ performance without violating the physical topology of the distribution, we must perform an exact conditional decomposition.

3.1 Basis Rotation

We rotate the Cartesian (T, q) space into a frame anchored to the vertical gradient vector. Let $\boldsymbol{g} = \nabla\phi$. We define the unit vectors parallel (u) and perpendicular (v) to the gradient:

$$\boldsymbol{e}_u = \frac{\boldsymbol{g}}{\|\boldsymbol{g}\|}, \quad \boldsymbol{e}_v = \frac{1}{\|\boldsymbol{g}\|}\begin{pmatrix} -g_q \\ g_T \end{pmatrix} \tag{3}$$

The transformation matrix is $M = [\boldsymbol{e}_u, \boldsymbol{e}_v]^T$. We project the turbulent covariance into this new frame:

$$\Sigma_{uv} = M\Sigma_{turb}M^T = \begin{pmatrix} s_u^2 & s_{uv} \\ s_{uv} & s_v^2 \end{pmatrix} \tag{4}$$

3.2 Conditional Decomposition (Rosenblatt Transformation)

Because $s_{uv} \neq 0$ generally, the turbulence along the path and perpendicular to the path are correlated. Assuming independence generates an un-tilted grid that fails to align with the joint probability mass. We utilize the conditional Rosenblatt transformation:

$$P(u, v) = P(v)P(u \mid v) \tag{5}$$

1. The Perpendicular Axis, $P(v)$: The marginal distribution perpendicular to the path is a pure 1D Gaussian, $\mathcal{N}(0, s_v^2)$. We discretize this axis using standard mapped quadrature to obtain N nodes $\{v_j\}$ and weights $\{w_{v,j}\}$.

2. The Parallel Axis, $P(u \mid v)$: Given a specific position v_j on the perpendicular axis, the conditional turbulent distribution along the u-axis is a Gaussian with shifted mean and reduced variance:

$$\mu_{u|v_j} = \left(\frac{s_{uv}}{s_v^2}\right) v_j \quad \text{and} \quad s_{u|v}^2 = s_u^2 - \frac{s_{uv}^2}{s_v^2} \tag{6}$$

3.3 Analytical Convolution

The full distribution along the u-axis is the conditional turbulence $s \equiv s_{u|v}$ convolved with the uniform vertical path of length $L = H\|\boldsymbol{g}\|$. Its cumulative distribution function (CDF), $F(x)$, is analytically tractable:

$$F(x) = \frac{s}{L}\left[\mathcal{I}\left(\frac{x+L/2}{s}\right) - \mathcal{I}\left(\frac{x-L/2}{s}\right)\right] \tag{7}$$

where $\mathcal{I}(y) = y\Phi(y) + \phi(y)$, with Φ and ϕ representing the standard normal CDF and PDF, respectively.

4 High-Order Analytical Inversion

To evaluate the quadrature, we must find the roots of $F(x) = p_i$, where p_i are the target Legendre probabilities. Numerical iterative root-finders are computationally prohibitive for inline evaluation. We require a direct $\mathcal{O}(1)$ analytical formulation parameterized by the non-dimensional ratio $\eta = s/L$.

4.1 The Analytical Jacobian

To perform high-order Newton-class refinement, we require the explicit derivatives of the convolution. Let $u_1 = \frac{x+L/2}{s}$ and $u_2 = \frac{x-L/2}{s}$.

$$F(x) = \frac{s}{L}\left[\mathcal{I}(u_1) - \mathcal{I}(u_2)\right] \quad \text{(CDF)} \tag{8}$$

$$f(x) = \frac{1}{2L}\left[\operatorname{erf}\left(\frac{u_1}{\sqrt{2}}\right) - \operatorname{erf}\left(\frac{u_2}{\sqrt{2}}\right)\right] \quad \text{(PDF)} \tag{9}$$

$$f'(x) = \frac{1}{Ls\sqrt{2\pi}}\left[e^{-u_1^2/2} - e^{-u_2^2/2}\right] \quad \text{(PDF Gradient)} \tag{10}$$

$$f''(x) = \frac{1}{Ls^2\sqrt{2\pi}}\left[-u_1 e^{-u_1^2/2} + u_2 e^{-u_2^2/2}\right] \quad \text{(PDF Curvature)} \tag{11}$$

4.2 Logistic Warping Seed Generation

Standard blending functions fail because the transition from a Box distribution ($\eta \to 0$) to a Gaussian ($\eta \to \infty$) is highly non-linear. Cornish-Fisher expansions diverge for small η due to the rigid boundaries of the uniform distribution.

The transition region ($\eta \in [0.05, 0.5]$) closely aligns with a Logistic distribution. By mapping the total variance to the scale parameter of a Logistic CDF, we derive an analytically stable initial guess x_0:

$$\mathrm{Var}_{tot} = s^2 + \frac{L^2}{12}, \quad S_{log} = \frac{\sqrt{3 \cdot \mathrm{Var}_{tot}}}{\pi}, \quad x_0 = S_{log} \ln\left(\frac{p_i}{1-p_i}\right) \tag{12}$$

4.3 Householder's Method (4th-Order Snapping)

For maximum computational efficiency, we seek to reach machine precision ($\sim 10^{-16}$) in a single step. Given the Logistic seed x_0, we snap the node using Householder's method, which utilizes the third derivative (f''). Let $h = \frac{p_i - F(x_0)}{f(x_0)}$:

$$x_{final} = x_0 + h\left[1 + \frac{1}{2}h\frac{f'(x_0)}{f(x_0)} + \frac{h^2}{6}\left(\left(\frac{f'(x_0)}{f(x_0)}\right)^2 + \frac{f''(x_0)}{f(x_0)}\right)\right] \tag{13}$$

5 Spectral Coordinate Mapping (Offline Optimization)

Transcendental snapping requires evaluating erf and exp. For CPU architectures where pure arithmetic is preferred, we map the non-linear domain to a spectral basis. We transform $\eta \to \xi \in [-1, 1]$ via logarithmic scaling:

$$\xi = \frac{2\log_{10}(\eta) - (\log_{10}\eta_{min} + \log_{10}\eta_{max})}{\log_{10}\eta_{max} - \log_{10}\eta_{min}} \tag{14}$$

5.1 Chebyshev Orthogonal Expansion

We project the true roots onto a Chebyshev polynomial basis $T_k(\xi)$. By the equioscillation principle, this minimizes the maximum error across the domain, avoiding the Runge's phenomenon inherent in standard polynomial or Padé fits.

$$x(\eta) \approx L \sum_{k=0}^{D} c_k T_k(\xi) \tag{15}$$

5.2 Barycentric Rational Interpolation

For strictly guaranteed stability without oscillations, Barycentric Rational Interpolation (Floater-Hormann) is employed over a set of control points ξ_k and true nodes y_k:

$$x(\xi) = L \left(\sum_{k=0}^{M} \frac{w_k}{\xi - \xi_k} y_k \right) \Bigg/ \left(\sum_{k=0}^{M} \frac{w_k}{\xi - \xi_k} \right) \tag{16}$$

where $w_k = (-1)^{k-d} \sum_{i=\max(0,k-d)}^{\min(k,M-d)} \prod_{j=i, j\neq k}^{i+d} \frac{1}{|\xi_k - \xi_j|}$ are statically determined weights.

6 Computational Architecture and Caching Strategies

For high-performance compiled implementations (e.g., Julia, Fortran):

- **Pointwise Math (Householder-4):** Evaluates the Logistic seed and analytical derivatives entirely inline. Highly favored by GPU architectures with Fused Multiply-Add (FMA) pipelines, requiring zero memory fetches and avoiding thread-divergence.
- **Spectral Coefficients (Chebyshev/Barycentric):** Reduces transcendental function calls to pure arithmetic. Requires caching a small array of coefficients ($N_{nodes} \times$ degree), which easily fits within CPU L1 cache.

7 Algorithm Assembly

Algorithm 1 Conditional Sheared $\mathcal{O}(N^2)$ Quadrature

Compute M, M^{-1}, and $\Sigma_{uv} = M\Sigma_{turb}M^T$.
Extract $s_v^2 = \Sigma_{uv}[1,1]$, $s_{uv} = \Sigma_{uv}[0,1]$.
Compute conditional variance $s_{u|v}^2 = s_u^2 - s_{uv}^2/s_v^2$.
Generate N mapped nodes $\{v_j, w_{v,j}\}$ for $\mathcal{N}(0, s_v^2)$.
Generate N base Pill nodes $\{u_{base,i}, w_{u,i}\}$ for $F_{pill}(x; L, s_{u|v})$ via Householder-4 or Chebyshev expansion.
$q_c \leftarrow 0$
for $j = 1$ to N **do**
  $\mu_{u|v_j} \leftarrow (s_{uv}/s_v^2)v_j$
  for $i = 1$ to N **do**
    $u_{ij} \leftarrow u_{base,i} + \mu_{u|v_j}$
    $\boldsymbol{x}_{ij} \leftarrow \boldsymbol{\mu} + M^{-1}(u_{ij}, v_j)^T$
    $f_{ij} \leftarrow \max(0, \boldsymbol{x}_{ij}^{(q)} - q_{sat}(\boldsymbol{x}_{ij}^{(T)}))$
    $q_c \leftarrow q_c + f_{ij} \cdot w_{u,i} \cdot w_{v,j}$
  end for
end for
return q_c